

Fair Test Planner: Change One Thing

BUILD • Science • 40 minutes

I can plan a safe fair comparison by naming one changed variable, one measured result, the controls and the repeats.

Prepare before pupils arrive

Setup: Prepare dry plan cards and keep W14 final sample codes/results inaccessible. Separately pretest one clearly labelled practice rig or the fixed pilot records.

Safety: Do not improvise with unknown or unmatched rock pieces. If a stable sealed practice rig is unavailable, approve the fixed pilot-record route as complete participation.

Equipment: Control cards, diagram of staff-built sealed rig, method planner, blank final results table, pilot checklist and pencil or communication aid.

Safety: Planning uses dry cards only. Pupils do not cut, seal, build, lift, scratch or prepare rigs or samples. Physical piloting happens only after local risk assessment, staff pretesting and teacher approval.

Fallback: Use large CHANGE / MEASURE / SAME / PILOT cards while an adult scribes the pupil's exact plan and selected access route.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Plan a simple fair test on rocks; pilot the method ready for the full investigation.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Last lesson: what method question did your audience bank? Last lesson: why did irrelevant evidence not help the choice?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	variable — a factor that can change control — keep this the same repeat — do the same method again Write the W14 question: Which supplied coded sample lets more water through in 120 seconds under this method? Change only the sample code.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Plan the W14 repeats before seeing final results. Use staff-prepared dry matched pieces and preserve every valid raw value.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each plan card to CHANGE, MEASURE, KEEP THE SAME or REPEAT / PILOT / SAFETY. Read the factor or action.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Produce a W14-ready fair-test plan and a separate W13B pilot checklist, leaving all final sample result cells blank. Record: W14 question, prediction, changed variable, measure with unit, at least three controls, repeat plan, blank results table, W13B pilot checks and safety stop.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain why W13B tests the method but does not answer the final A/B question. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Change sample code; measure water collected in mL; keep volume, time, area and thickness the same.

Correct. The sample is the planned difference, collected water is the measure and the important method factors are controlled.

Misconception repair

Repaired. Fairness is about the controlled comparison, and the pilot checks the method rather than answering the final question.

Guided checkpoint

Correct. The sample is the planned difference, collected water is the measure and the important method factors are controlled. Choose the plan where both sample conditions receive the same water method.

Sample code

CHANGE. Change: sample code is the one planned difference in the full comparison. Which factor is deliberately different between conditions?

Collected water

MEASURE. Measure: collected-water volume is the outcome for each trial. Look for the result recorded with a unit.

Starting volume

KEEP THE SAME. Control: equal starting volume prevents one sample receiving more water. Starting volume is not the planned changed variable.

Test time

KEEP THE SAME. Control: each sample has the same time for water passage. Both conditions need the same timed window.

Exposed area

KEEP THE SAME. Control: equal exposed area keeps the water route comparable. A wider opening would create a different route area.

Prepared thickness

KEEP THE SAME. Control: water should not have a much longer path through one condition. Thickness affects the route length and should not change with sample code.

Final repeats

REPEAT / PILOT / SAFETY. Repeat plan: repeated valid records will show pattern and variation in W14. This action occurs again after the method is approved.

Practice pilot

REPEAT / PILOT / SAFETY. Pilot: it checks whether the method works without answering which final sample is more permeable. The pilot tests the procedure before final A/B evidence is collected.

Safety stop

REPEAT / PILOT / SAFETY. Safety gate: the method states when the physical route must stop. A complete investigation plan includes the stop rule.

Model limitation

A fair plan reduces alternative explanations but cannot remove natural sample variation. The pilot can improve procedure, yet only W14's approved repeated comparison may address the investigation question.

Independent evidence

W14 question, prediction, changed variable, measure with unit, at least three controls, repeat plan, blank results table, W13B pilot checks and safety stop.

Supported: Build the method from fixed picture cards and choose one pilot check and one safety stop. Use one card at a time: Change ___. Measure ___. Keep ___ the same. Pilot ___.

Standard: Write the full plan, justify three controls and prepare separate pilot and final-data sheets. Check that no final A/B result or conclusion appears before W14.

Stretch: Evaluate a flawed alternative, justify thickness and exposed-area controls, and explain what the pilot can and cannot establish. Separate procedural validity from the final scientific comparison.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W13A_Fair_Test_Planner_Change_One_Thing_Explore.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.